

**AFFILIATED TO
VEER NARMAD SOUTH GUJARAT UNIVERSITY, SURAT**

PROJECT REPORT

**AS A PARTIAL REQUIREMENT FOR THE DEGREE OF
BACHELOR OF COMPUTER APPLICATIONS (BCA)**

**BCA SEM V
A.Y. 2026-27**

PROJECT REPORT ON “ONLINE LIBRARY MANAGEMENT SYSTEM”

Submitted By

Name: _____

Exam No.: _____

Roll No.: _____

Project Guide: _____

Acknowledgement

The successful completion of the Online Library Management System required guidance, support and encouragement from many people. We are grateful to the management, Principal, IQAC coordinator, Head of the BCA Department, project guide and teaching staff of SDJ International College, Vesu, for providing the academic environment and direction required to complete this project.

We sincerely thank our project guide for valuable suggestions during requirement analysis, design, implementation and testing. We also thank the teaching and non-teaching staff of the IT Department for their timely support. Finally, we thank our classmates, friends and family members for their encouragement and feedback.

Student Name (Exam Seat Number): _____

Student Name (Exam Seat Number): _____

Student Name (Exam Seat Number): _____

Declaration

We hereby declare that the project entitled “Online Library Management System” is submitted as a partial requirement for the Bachelor of Computer Applications (BCA), Semester V, A.Y. 2026-27. The project is developed for academic purposes using PHP, MySQL, HTML, CSS, JavaScript, jQuery and AJAX.

Particular	Details
Project Title	Online Library Management System
Course	BCA
Semester	V
Academic Year	2026-27
University	Veer Narmad South Gujarat University, Surat
Front End	HTML, CSS, JavaScript, jQuery, AJAX
Back End	PHP 7.x / 8.x
Database	library_db (MySQL / MariaDB)
Server	XAMPP (Apache + MySQL)
Application Type	Web-based library management system with book issue, return and book selling

Student Signature: _____

Date: _____

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1 Introduction

1.1 Project Description

The Online Library Management System is a web-based application that manages the daily work of a college library. It keeps records of books, categories, authors and students in one MySQL database, and it replaces paper registers with simple web pages.

The student side supports registration (a unique Student ID is generated), login, viewing the listed books with cover pictures, checking issued books with their return date and time, updating the profile, and changing or recovering the password. Students can also buy their own copy of a book from the built-in Book Store using a cart and checkout.

The admin side provides a dashboard and full management of categories, authors and books, issuing a book to a student, updating return details with fine, searching a student by Student ID, and processing book orders.

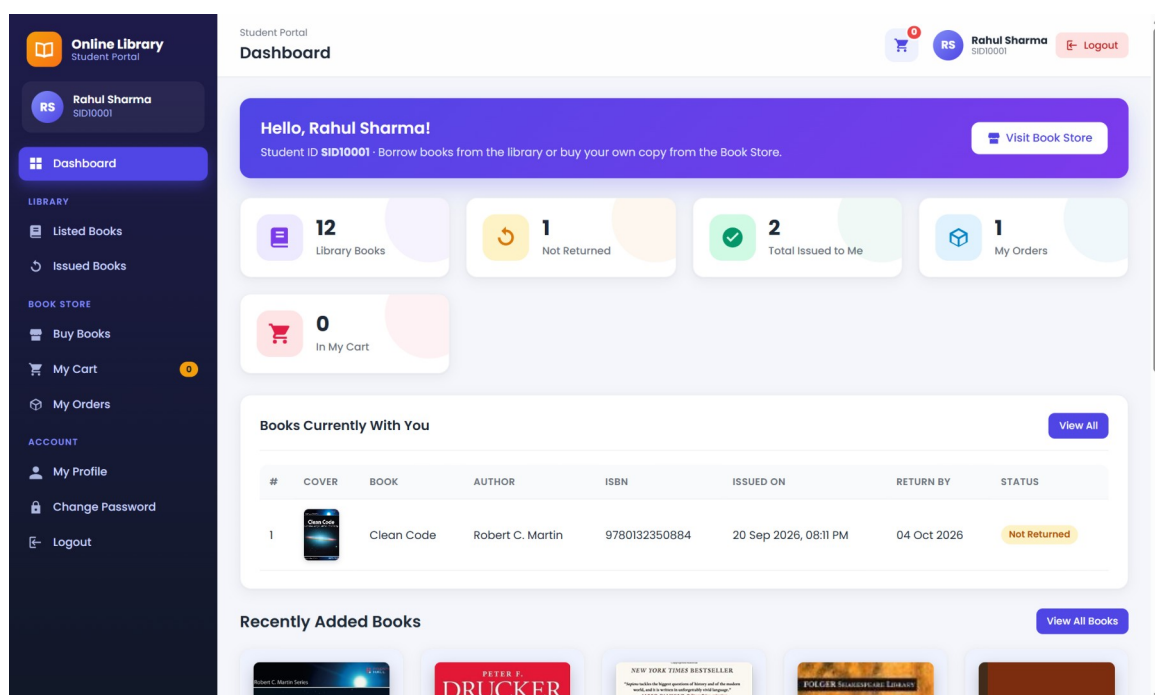


Figure 1.1: Student dashboard of the Online Library Management System

1 Introduction

1.2 Project Profile

The project is named Online Library Management System. It has two main modules, Student and Admin, and an additional Book Store (book selling) module. The fine for a late return is Rs. 5 per day and the default issue period is 14 days.

The project database is named library_db. The local URL is <http://localhost/library/> for students and <http://localhost/library/admin/> for the administrator. Demo accounts are provided for testing only and should be changed before any public use.

Item	Details
Project Name	Online Library Management System
Modules	Student, Admin, Book Store
Database	library_db (9 tables)
Student URL	http://localhost/library/
Admin URL	http://localhost/library/admin/
Fine / Issue period	Rs. 5 per late day / 14 days
Payment options (Book Store)	Cash on Delivery, Pay at Library Counter
Supported Browsers	Google Chrome, Mozilla Firefox, Microsoft Edge, Opera

2 Environment Description

2.1 Hardware and Software Requirements

The application runs on a normal computer that can run a local web server, PHP and MySQL. A computer with 4 GB RAM is enough for development, while 8 GB is better when the editor, browser and server run together.

Type	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB (8 GB recommended)
Hard Disk	1 GB free space for XAMPP and project
Operating System	Windows 10 / Windows 11 (also Linux / macOS)
Server Software	XAMPP / WAMP / MAMP / LAMP (any one)
PHP	PHP 7.x or PHP 8.x
Database	MySQL / MariaDB with phpMyAdmin
Browser	Chrome, Firefox, Edge or Opera
Editor	Visual Studio Code or any code editor

2 Environment Description

2.2 Technologies Used

PHP is used for server-side processing, sessions, login and database work. The MySQLi extension with prepared statements connects PHP to MySQL, and passwords are stored using bcrypt hashing (password_hash).

HTML and CSS build the pages, and JavaScript with jQuery adds interactivity. AJAX is used for live book search, student and book lookup on the Issue Book page, student search, and cart updates without reloading the page.

Technology	Use in the Project
HTML5 / CSS3	Page structure, responsive layout, sidebar, cards
JavaScript / jQuery	Menus, form checks, cart actions, image preview
AJAX	Live search, student/book lookup, cart update
PHP	Business logic, sessions, file upload, validation
MySQL / MariaDB	Stores books, students, issues, carts and orders
XAMPP	Apache web server + MySQL + phpMyAdmin

3 System Analysis and Planning

3.1 Existing System and its Drawbacks

In a manual library, the librarian writes book issue and return details in registers. Students must visit the library to know which books are available, and fines are calculated by hand.

- Searching for a book or a student record takes a lot of time.
- Registers can be lost, damaged or filled with mistakes.
- Students cannot check their issued books or return dates from home.
- Late fines are calculated manually and may be wrong.
- There is no easy way for students to buy a copy of a book.

The Online Library Management System solves these problems with a central database and separate Student and Admin panels. It is an academic project for digitalising a college library.

3 System Analysis and Planning

3.2 Feasibility Study

Technical feasibility: The system uses common and free technologies (PHP, MySQL, HTML, CSS, JavaScript) and runs on a local XAMPP server.

Economic feasibility: All development tools are open source, so there is no software cost. It also saves paper and staff time.

Operational feasibility: The pages are simple, with clear menus for students and the admin. Very little training is needed.

Schedule feasibility: The project is divided into small modules (authentication, books, issue/return, book store), so it can be completed within the semester.

3 System Analysis and Planning

3.3 Requirement Gathering and Analysis

Requirements were collected from the college library working process and the given project list.

Functional requirements: student registration with Student ID, login, password recovery, listed books, issued books with return date-time, profile update and change password. Admin requirements are category, author and book management, issue and return of books, student search and details, and change password.

Additional requirements: book cover pictures, a Book Store where students can buy books through a cart and checkout, and order processing by the admin.

Non-functional requirements: secure passwords, fast pages, simple design, and support for all modern browsers.

4 Proposed System

4.1 Scope

The scope covers the full library process from student registration to book issue, return and fine. Book records include name, category, author, ISBN, price, copies for sale and a cover picture.

The system also covers a small Book Store: cart records, orders, order items, stock control and order status (Pending, Confirmed, Dispatched, Delivered, Cancelled). Online card payment is not part of the scope.

4 Proposed System

4.2 Project Modules

Major modules are: Student, Admin, Book Store and Common/System.

- **Student:** register, login/logout, forgot password, dashboard, listed books, issued books, my profile, change password.
- **Admin:** dashboard, categories, authors, books (with picture), issue book, manage issued books, return with fine, registered students, search student, change password.
- **Book Store:** buy books, cart, checkout, my orders, cancel order (student) and book orders management (admin).
- **Common/System:** database connection, sessions, helper functions, AJAX endpoints and layout.

4 Proposed System

4.3 Module Objectives / Functionalities / Constraints

The student module gives students a simple way to find books and track what they have borrowed. The admin module puts all library control in one place. The Book Store module lets students buy books without visiting the shop.

Module	Objective	Constraint
Student	Self-service access to books and issue history	Login required; blocked students cannot log in
Admin	Manage catalogue, issues, returns, students	Only admin can access admin pages
Issue / Return	Track loans and fines	A book can be issued to only one student at a time
Book Store	Sell copies of books	Cannot buy more than stock or 5 copies of one book

Payment in the Book Store is Cash on Delivery or Pay at Library Counter. It is not a live online banking or card gateway.

5 Detail Planning

5.1 DFD / UML – Use Case & Activity Flow

The high-level flow contains Student, Admin, the web application and the MySQL database. Requests from students and the admin are processed by PHP pages and stored in MySQL.

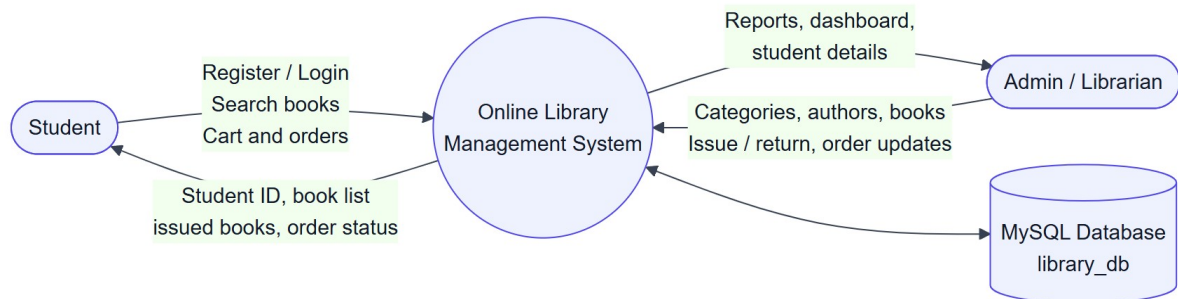


Figure 5.1: Context level (Level 0) Data Flow Diagram

The Level 1 DFD below shows the main processes and the tables each one uses.

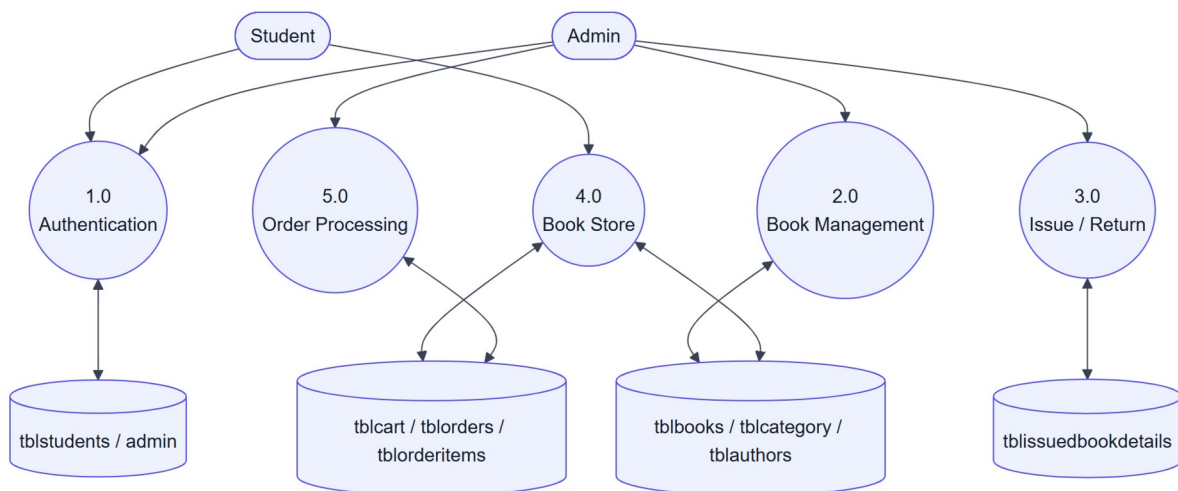


Figure 5.2: Level 1 Data Flow Diagram

5 Detail Planning

5.1 Use Case Diagram

Students and the admin use different parts of the system. Both can log in and recover or change the password.

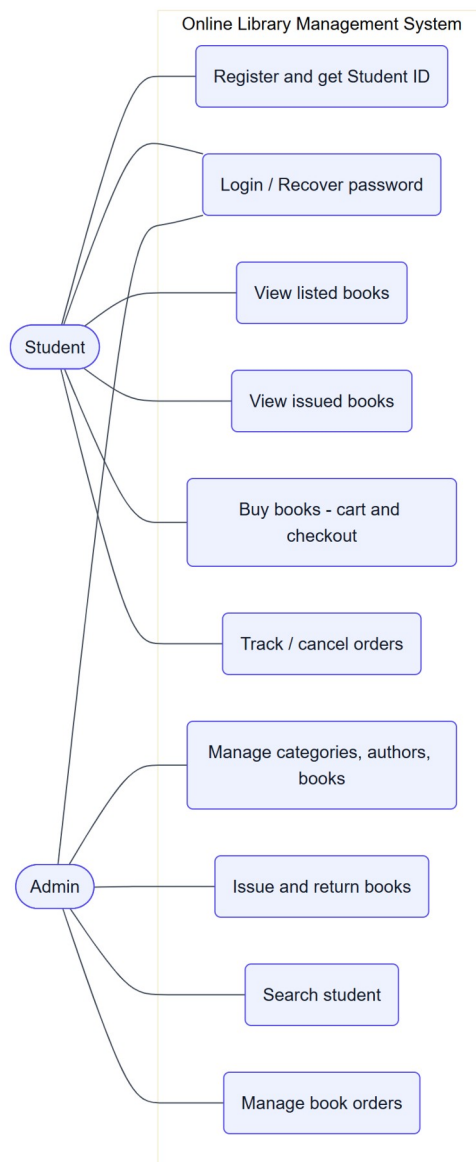


Figure 5.3: Use case diagram

5 Detail Planning

5.1.1 Student Activity Flow

Student opens the website → registers (gets Student ID) or logs in → opens the dashboard → views listed books or issued books → buys books through Buy Books → adds to cart → checks out → order is placed.

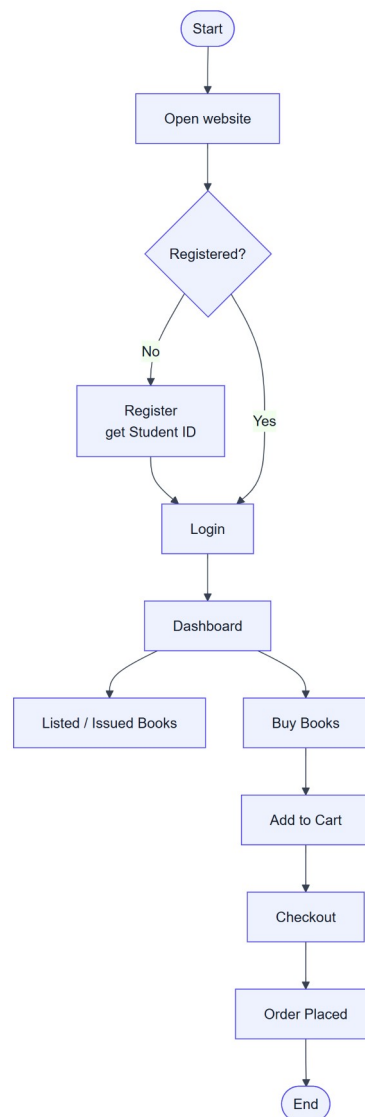


Figure 5.4: Student activity diagram

5 Detail Planning

5.1.2 Admin Activity Flow

Admin logs in → opens dashboard → manages categories, authors and books → issues a book to a student → marks the book as returned with fine → processes book orders (confirm, dispatch, deliver or cancel).

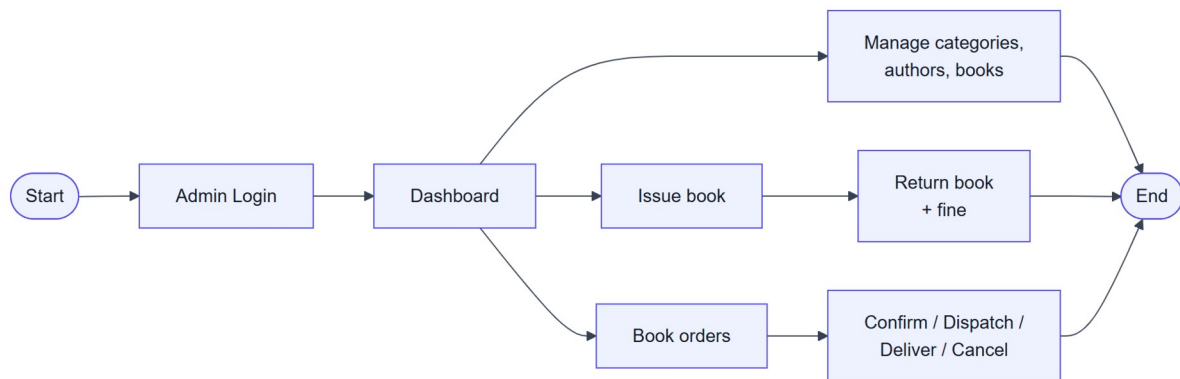


Figure 5.5: Admin activity diagram

5 Detail Planning

5.2 Process Specification

5.2.1 Process Specification – Authentication

Registration creates a student record with a unique Student ID (for example SID10001). Login checks the Student ID or email and password and creates a session. The admin has a separate login page. Logout ends the session.

Forgot password checks the registered email and mobile number and lets the student set a new password. All passwords are stored as bcrypt hashes.

Input	Process	Output
Name, email, mobile, password	Validate, check duplicate email, create Student ID	Account created + Student ID
Student ID / email + password	Verify password hash and status	Session started, dashboard
Email + mobile + new password	Match record and update hash	Password reset

5 Detail Planning

5.2.2 Process Specification – Book Management

Books are stored in the tblbooks table and linked to a category and an author. Admin pages support add, edit and delete. Book data includes name, ISBN, price, copies for sale and a cover picture (JPG, PNG, WEBP or GIF up to 2 MB).

A category or author that still has books cannot be deleted, and a book that is currently issued cannot be deleted.

5 Detail Planning

5.2.3 Process Specification – Issue and Return

The admin types a Student ID and a book ISBN. AJAX shows the student name and whether the book is available. The book is then issued with a return date (default 14 days).

When the book is returned, the system calculates the fine as overdue days $\times$ Rs. 5, saves the return date and time, and marks the record as returned.

5 Detail Planning

5.2.4 Process Specification – Cart and Checkout

Cart items are stored in tblcart with Student ID, book and quantity. Adding, changing and removing items is done with AJAX, and the stock and the 5-copy limit are checked each time.

Checkout collects the payment method, address (for Cash on Delivery) and contact number. The order is saved in a database transaction: stock is checked again, the order and order items are created, stock is reduced and the cart is emptied.

5 Detail Planning

5.2.5 Process Specification – Order Processing

Orders are stored in `tblorders` and `tblorderitems`. The admin changes the status from Pending to Confirmed, Dispatched and Delivered, and can add a remark for the student.

Delivered marks the payment as Paid. Cancelled (by the admin, or by the student before dispatch) returns the copies to stock. Delivered and Cancelled orders are final.

5 Detail Planning

5.3 Entity-Relationship Diagram

5.3.1 Database Relationships

The database library_db contains 9 tables: admin, tblcategory, tblauthors, tblbooks, tblstudents, tblissuedbookdetails, tblcart, tblorders and tblorderitems.

Parent Table	Child Table	Relationship
tblcategory	tblbooks	One category has many books
tblauthors	tblbooks	One author has many books
tblstudents	tblissuedbookdetails	One student borrows many books
tblbooks	tblissuedbookdetails	One book has many issue records
tblstudents	tblcart / tblorders	One student has cart items and orders
tblorders	tblorderitems	One order contains many items

5 Detail Planning

5.3 Entity-Relationship Diagram

The ERD shows the main entities and their relationships. Books belong to categories and authors, and appear in issue records, carts and order items.

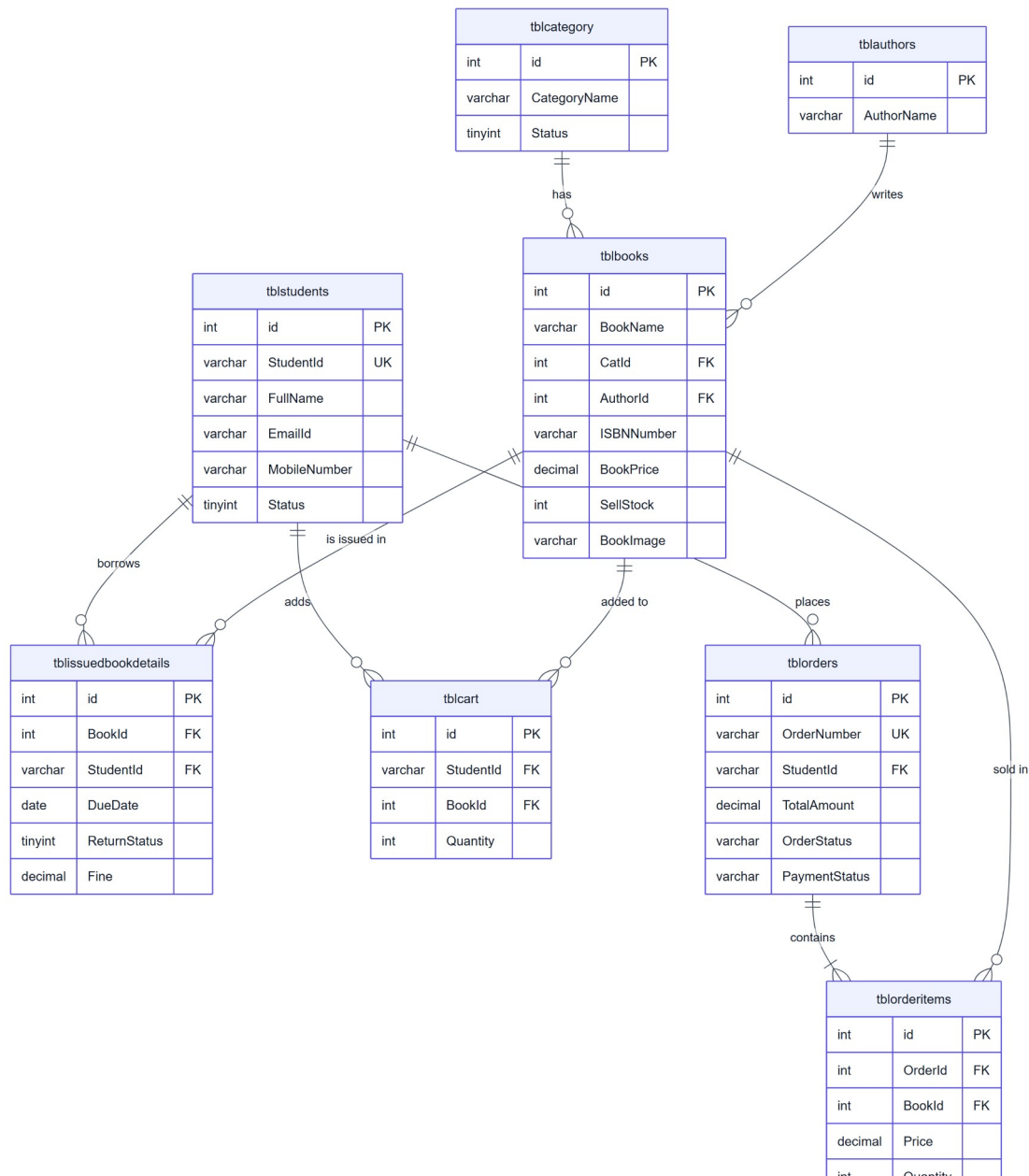


Figure 5.6: Entity-Relationship Diagram of library_db

6 System Design

6.1 Database Design – User and Role Tables

The admin table stores the administrator login. The tblstudents table stores student accounts. Both store passwords as bcrypt hashes.

Table: admin

Field	Type	Description
id	INT, PK	Admin ID
FullName	VARCHAR(100)	Admin name
AdminEmail	VARCHAR(120)	Email
UserName	VARCHAR(50), UNIQUE	Login username
Password	VARCHAR(255)	Password hash

Table: tblstudents

Field	Type	Description
id	INT, PK	Row ID
StudentId	VARCHAR(20), UNIQUE	Generated ID e.g. SID10001
FullName	VARCHAR(120)	Student name
EmailId	VARCHAR(120), UNIQUE	Email
MobileNumber	VARCHAR(15)	10-digit mobile
Password	VARCHAR(255)	Password hash
Status	TINYINT	1 = active, 0 = blocked
RegDate	TIMESTAMP	Registration date

6 System Design

6.1 Database Design – Book Tables

tblcategory and tblauthors are master tables. tblbooks stores each book and links to a category and an author.

Table: tblcategory

Field	Type	Description
id	INT, PK	Category ID
CategoryName	VARCHAR(150)	Name
Status	TINYINT	1 = active

Table: tblauthors

Field	Type	Description
id	INT, PK	Author ID
AuthorName	VARCHAR(150)	Author name

Table: tblbooks

Field	Type	Description
id	INT, PK	Book ID
BookName	VARCHAR(200)	Title
CatId	INT, FK	Category
AuthorId	INT, FK	Author
ISBNNumber	VARCHAR(30), UNIQUE	ISBN
BookPrice	DECIMAL(10,2)	Price
SellStock	INT	Copies for sale
BookImage	VARCHAR(255)	Cover picture file

6 System Design

6.1 Database Design – Issue Table

tblissuedbookdetails stores every book issue. The ReturnStatus field shows whether the book has come back, and Fine stores the amount collected.

Table: tblissuedbookdetails

Field	Type	Description
id	INT, PK	Issue ID
BookId	INT, FK	Issued book
StudentId	VARCHAR(20), FK	Student
IssueDate	TIMESTAMP	Issue date and time
DueDate	DATE	Return by date
ReturnDate	TIMESTAMP	Actual return date and time
ReturnStatus	TINYINT	0 = not returned, 1 = returned
Fine	DECIMAL(10,2)	Fine collected

6 System Design

6.1 Database Design – Cart and Order Tables

tblcart holds books a student wants to buy. tblorders stores the order and tblorderitems stores each book in the order with the price at the time of buying.

Table: tblcart

Field	Type	Description
id	INT, PK	Row ID
StudentId	VARCHAR(20), FK	Student
BookId	INT, FK	Book
Quantity	INT	Copies

Table: tblorders

Field	Type	Description
id	INT, PK	Order ID
OrderNumber	VARCHAR(30), UNIQUE	e.g. ORD2609251234
StudentId	VARCHAR(20), FK	Student
TotalAmount	DECIMAL(10,2)	Order total
PaymentMethod	VARCHAR(50)	COD / Library Counter
PaymentStatus	VARCHAR(20)	Unpaid / Paid
OrderStatus	VARCHAR(20)	Pending ... Delivered / Cancelled

Table: tblorderitems

Field	Type	Description
id	INT, PK	Row ID
OrderId	INT, FK	Order
BookId	INT, FK	Book
Price	DECIMAL(10,2)	Price when ordered
Quantity	INT	Copies

6 System Design

6.1.1 Database Table Summary

The database tables and their purposes are summarised below.

Table	Purpose
admin	Administrator login
tblstudents	Student accounts and Student ID
tblcategory	Book category master
tblauthors	Author master
tblbooks	Book master with price, stock and cover
tblissuedbookdetails	Book issue, return and fine records
tblcart	Student shopping cart
tblorders	Book order header
tblorderitems	Books inside an order

6 System Design

6.1.2 Key Book Fields

The tblbooks table includes BookName, CatId, AuthorId, ISBNNumber, BookPrice, SellStock and BookImage.

ISBNNumber is unique, so the admin can issue a book by typing only its ISBN. SellStock is the number of copies that can be sold in the Book Store; 0 means the book is not for sale. BookImage stores the cover picture file name.

6 System Design

6.1.3 Key Issue and Order Fields

For issues, the dashboard and the student pages use IssueDate, DueDate, ReturnDate, ReturnStatus and Fine to show whether a book is Not Returned, Overdue or Returned.

For orders, the pages use OrderNumber, OrderDate, TotalAmount, PaymentMethod, PaymentStatus and OrderStatus. tblorderitems copies the book name, ISBN and price, so order history stays correct even if a book is later edited or deleted.

6 System Design

6.2 User Interface – Home Page and Login

The home page has a colourful banner with the Student Login form and shows the book collection with cover pictures and an Available / Issued label.

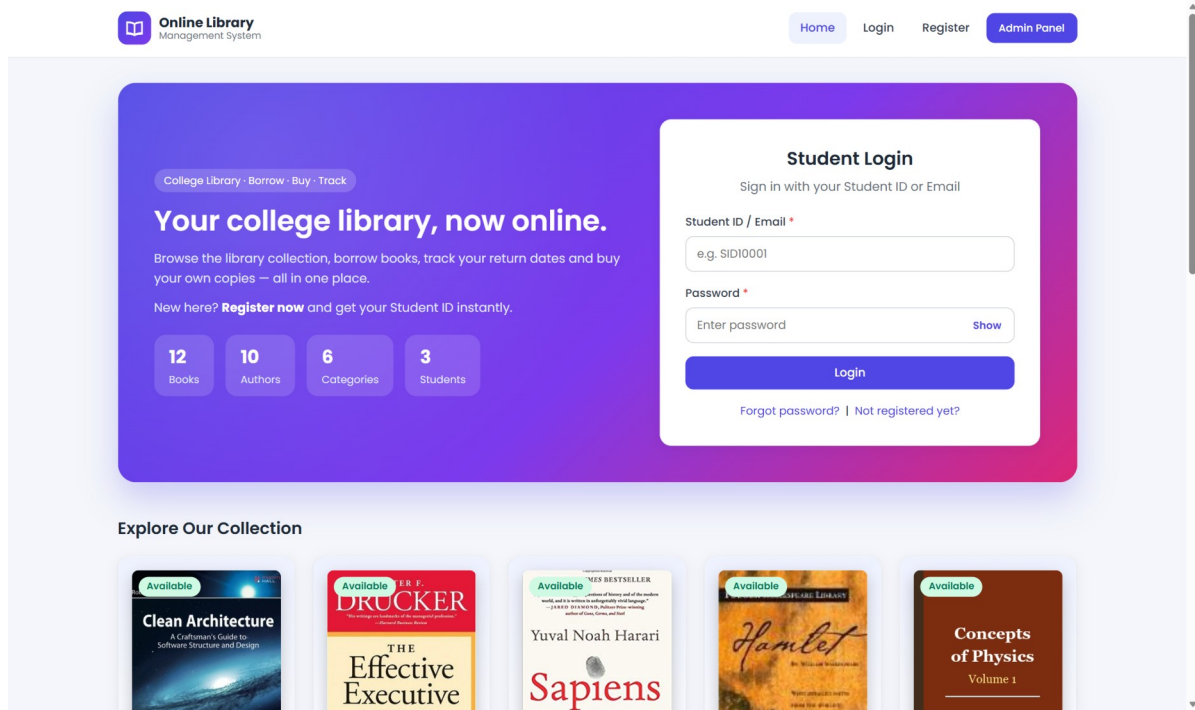


Figure 6.1: Home page with student login

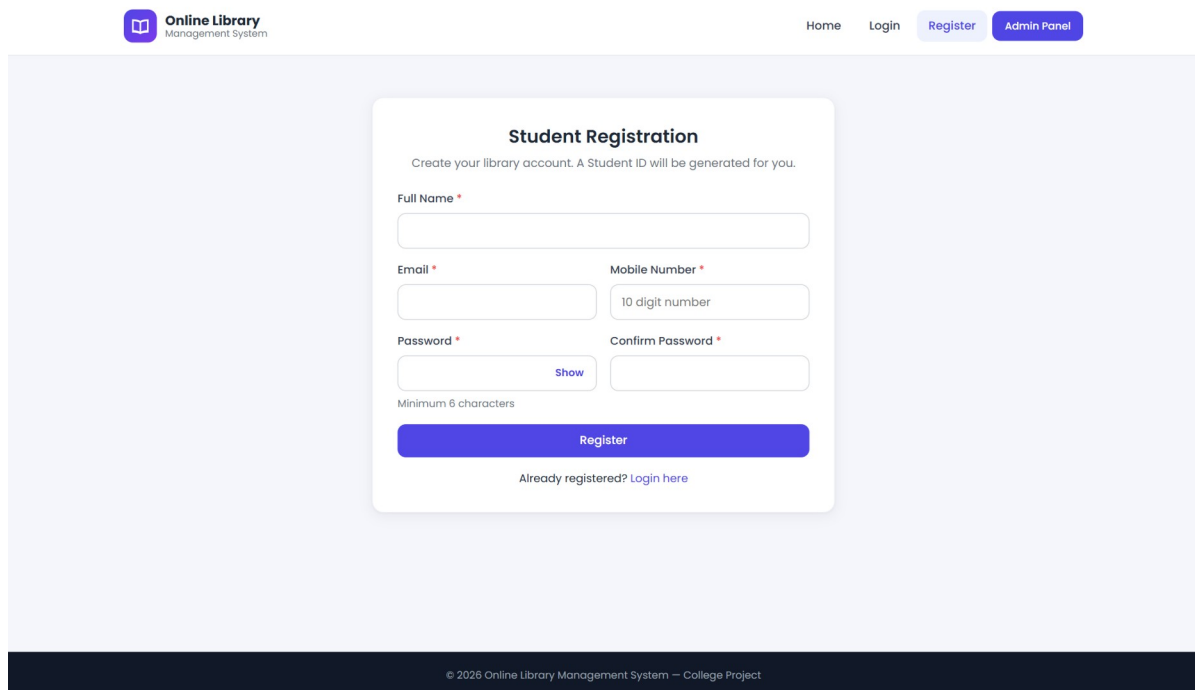


Figure 6.2: Student registration page

6 System Design

6.2 User Interface – Listed Books and Book Store

Listed Books shows every book as a card with its cover. The search box filters the cards instantly using AJAX. The Book Store shows price and stock and has an Add to Cart button.

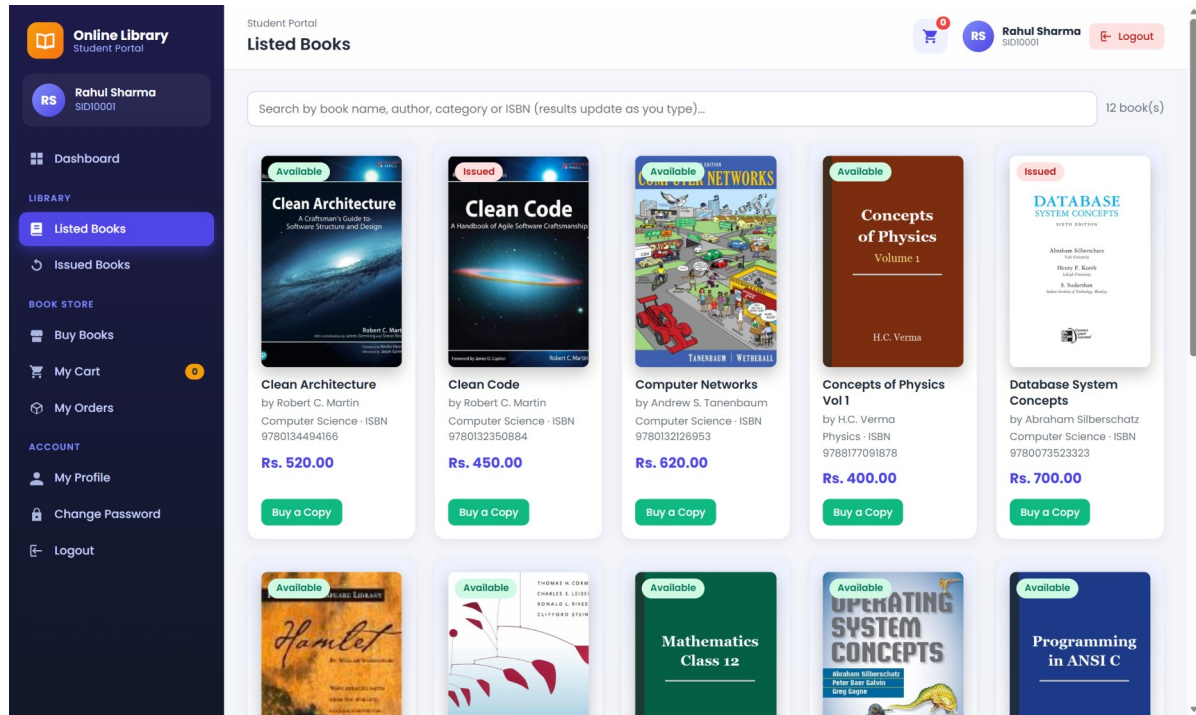


Figure 6.3: Listed books with live search

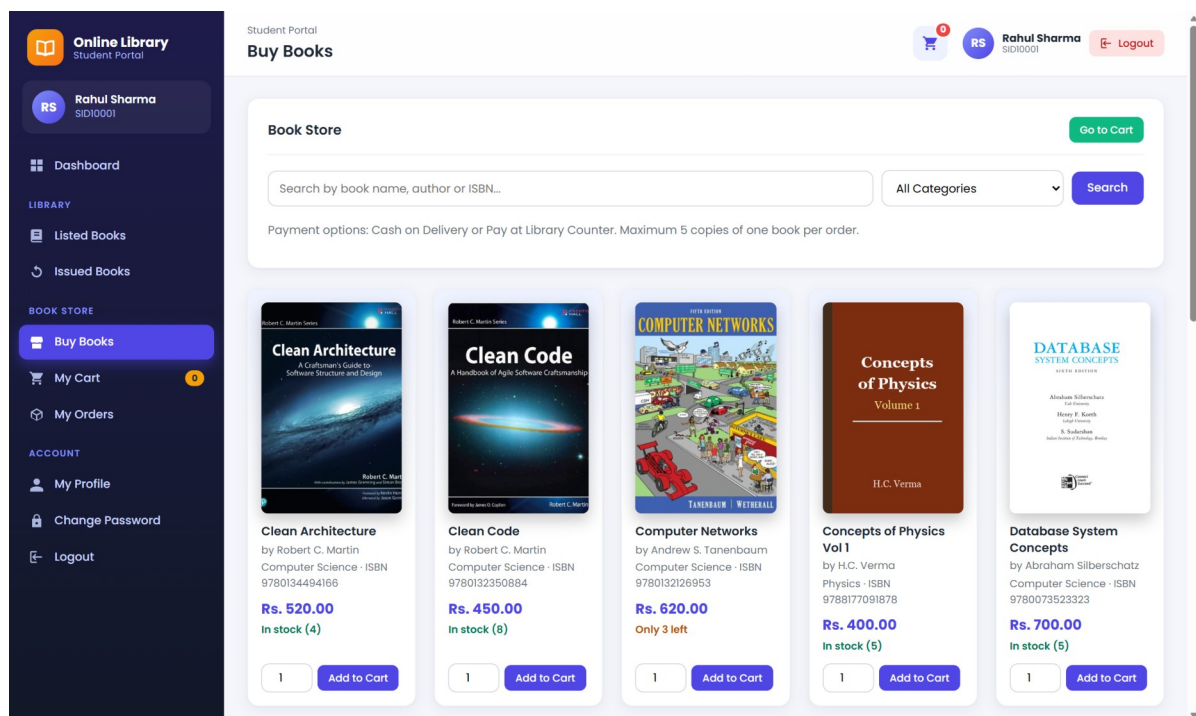


Figure 6.4: Book Store (Buy Books)

6 System Design

6.2 User Interface – Cart, Checkout and Orders

The cart updates quantities and totals without reloading. Checkout offers Cash on Delivery or Pay at Library Counter. The order page shows a progress bar.

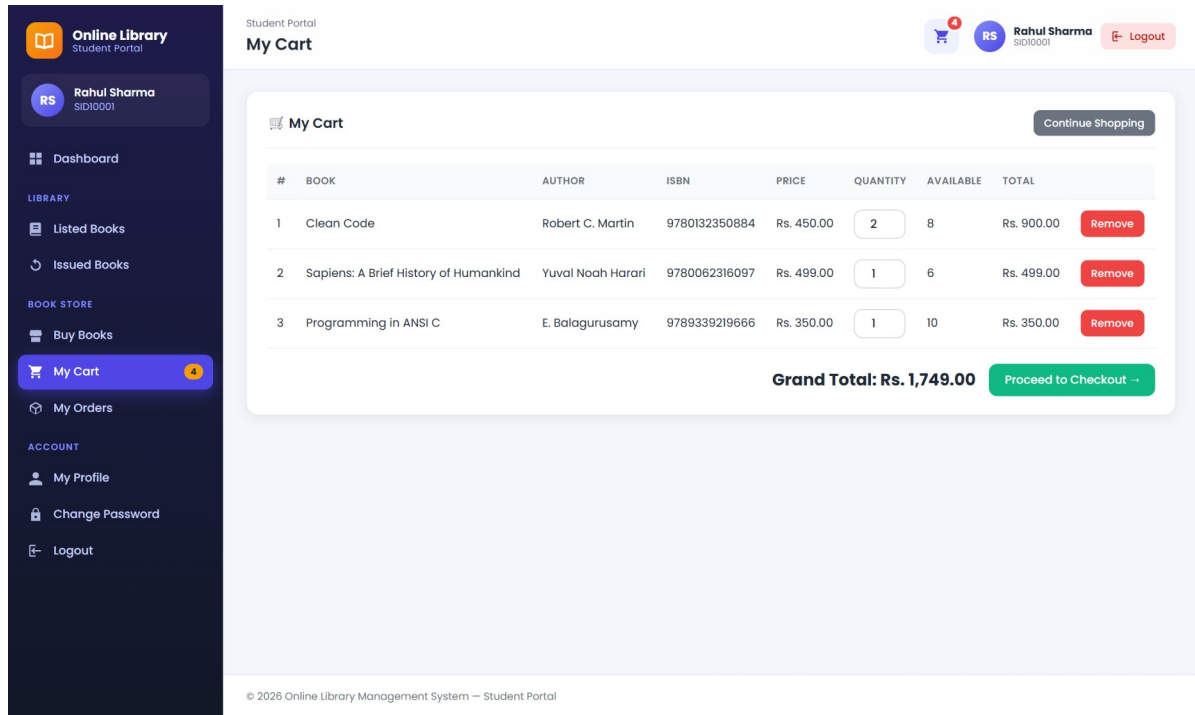


Figure 6.5: Shopping cart

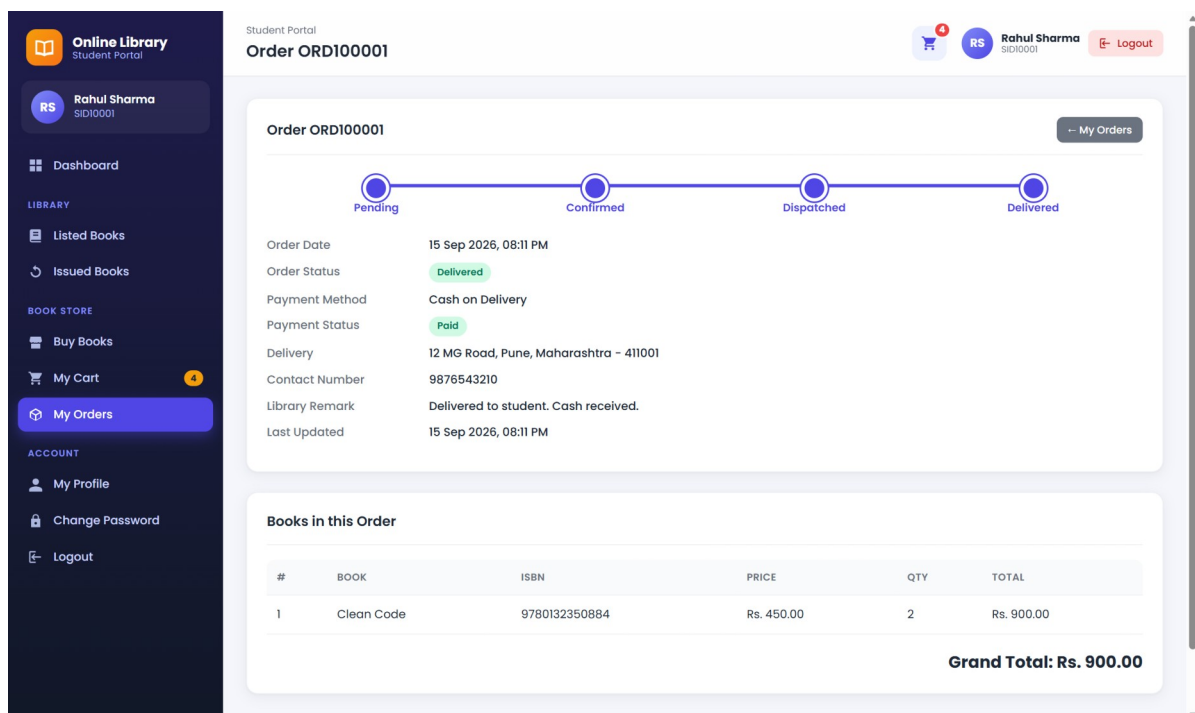


Figure 6.6: Order details with progress bar

6 System Design

6.2 User Interface – Issued Books

Issued Books shows the cover, issue date and time, due date, returned date and time, fine and status of every book issued to the student.

Student Portal

Issued Books

Books Issued to You

Fine: Rs. 5 per day after due date

#	COVER	BOOK	AUTHOR	ISBN	ISSUED DATE & TIME	DUE DATE	RETURNED DATE & TIME	FINE (RS.)	STATUS
1		Clean Code	Robert C. Martin	9780132350884	20 Sep 2026, 08:11 PM	04 Oct 2026	-	0.00	Not Returned
2		Hamlet	William Shakespeare	9780743477123	26 Aug 2026, 08:11 PM	09 Sep 2026	11 Sep 2026, 08:11 PM	10.00	Returned

© 2026 Online Library Management System – Student Portal

Figure 6.7: Issued books page (student)

Student Portal

Checkout

#	BOOK	ISBN	PRICE	QUANTITY	TOTAL
3	Programming in ANSI C	9789339219666	Rs. 350.00	1	Rs. 350.00

Grand Total: Rs. 1,749.00

Payment & Delivery

☒ Cash on Delivery
Books are delivered to your address. Pay cash when you receive them.

☐ Pay at Library Counter
Collect the books from the library counter and pay there.

Delivery Address *

House no, street, city, PIN code

Contact Number *

9876543210

Place Order (Rs. 1,749.00)

© 2026 Online Library Management System – Student Portal

Figure 6.8: Checkout page

6 System Design

6.2 User Interface – Admin Dashboard

The admin dashboard shows a welcome banner and stat cards for total books, books on loan, returned books, students, authors, categories, orders, pending orders and sales. It also lists recent issues and recent orders.

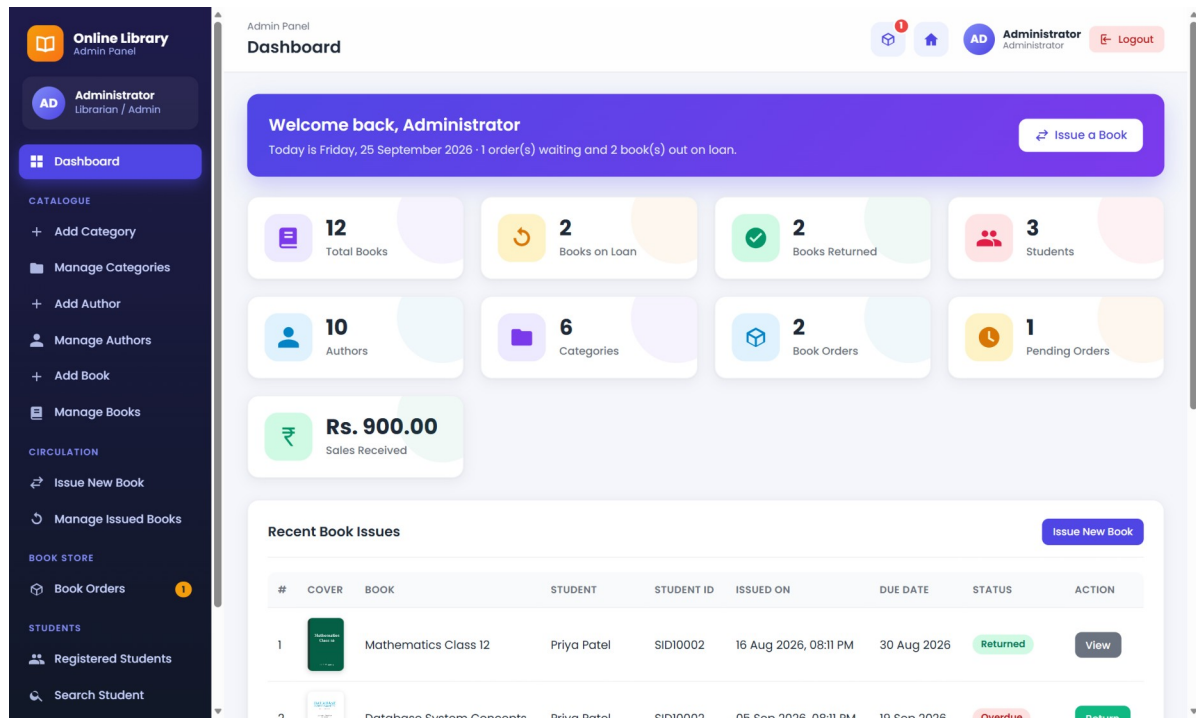


Figure 6.9: Admin dashboard

The Admin Login page is a simple, clean interface for administrators to access the system. It includes a header with navigation links and a central login form.

Admin Login Form:

- Username ***:
- Password ***: [Show](#)
- Login**:
- [Back to Student Website](#)

Footer: © 2026 Online Library Management System – College Project

Figure 6.10: Admin login

6 System Design

6.2 User Interface – Admin Modules

The sidebar groups admin pages into Catalogue, Circulation, Book Store and Students. Manage Books shows the cover, price, copies for sale and status with Edit and Delete buttons. Add Book includes a picture upload with preview.

Online Library Admin Panel

Administrator
Librarian / Admin

Dashboard

CATALOGUE

- + Add Category
- Manage Categories
- + Add Author
- Manage Authors
- + Add Book
- Manage Books**

CIRCULATION

- Issue New Book
- Manage Issued Books

BOOK STORE

- Book Orders

STUDENTS

- Registered Students
- Search Student

Admin Panel

Manage Books

All Books (12) [+ Add Book](#)

Filter by book, author, category, ISBN...

#	COVER	BOOK NAME	CATEGORY	AUTHOR	ISBN	PRICE (RS.)	COPIES FOR SALE	LIBRARY STATUS	ACTION
1		Clean Architecture	Computer Science	Robert C. Martin	9780134494166	520.00	4	Available	Edit Del
2		The Effective Executive	Management	Peter F. Drucker	9780060833459	380.00	Not for sale	Available	Edit Del
3		Sapiens: A Brief History of Humankind	History	Yuval Noah Harari	9780062316097	499.00	6	Available	Edit Del
4		Hamlet	Literature	William Shakespeare	9780743477123	250.00	2	Available	Edit Del
5		Concepts of Physics Vol 1	Physics	H.C. Verma	9788177091878	400.00	5	Available	Edit Del
6		Mathematics Class 12	Mathematics	R.D. Sharma	9789383182480	550.00	7	Available	Edit Del

Figure 6.11: Manage books

Online Library Admin Panel

Administrator
Librarian / Admin

Dashboard

CATALOGUE

- + Add Category
- Manage Categories
- + Add Author
- Manage Authors
- + Add Book**
- Manage Books

CIRCULATION

- Issue New Book
- Manage Issued Books

BOOK STORE

- Book Orders

STUDENTS

- Registered Students
- Search Student

Admin Panel

Add Book

Book Info [Manage Books](#)

Book Name *

Category * **Author ***

-- Select Category -- -- Select Author --

ISBN Number * **Price (Rs.) *** **Copies for Sale**

Book Picture

[Choose File](#) No file chosen

JPG, PNG, WEBP or GIF. Maximum 2 MB. Optional.

[Add Book](#)

© 2026 Online Library Management System – Admin Panel

Figure 6.12: Add book with picture upload

6 System Design

6.2 User Interface – Issue and Return

On Issue New Book, the student name and book name are filled automatically with AJAX. On return, the fine is calculated automatically.

Online Library Admin Panel

Issue New Book

Issue a Book to Student

Type the Student ID and the book ISBN. The student name and book name are fetched automatically (AJAX).

Student ID * Student Name

Student found: Priya Patel (priya@example.com)

Book ISBN Number * Book Name

Book found: Computer Networks by Andrew S. Tanenbaum – Available

Return Date *

Default is 14 days from today. Fine after this date: Rs. 5/day.

Issue Book

© 2026 Online Library Management System – Admin Panel

Figure 6.13: Issue new book with AJAX lookup

Online Library Admin Panel

Return Book

Update Return Details

Book Name: Database System Concepts

Author: Abraham Silberschatz

ISBN: 9780073523323

Student: Priya Patel (SID10002)

Email / Mobile: priya@example.com / 9123456780

Issued On: 05 Sep 2026, 08:11 PM

Due Date: 19 Sep 2026

Status: **Overdue by 6 day(s)**

Return Date & Time:

Fine (Rs.):

Auto calculated: 6 overdue day(s) x Rs. 5. You can change it.

Mark as Returned

© 2026 Online Library Management System – Admin Panel

Figure 6.14: Return book with fine

6 System Design

6.2 User Interface – Book Orders and Students

Book Orders lists all orders with filters by status. The admin can also search a student by Student ID and view the full profile and issue history.

Admin Panel

Manage Book Orders

Administrator Librarian / Admin

Dashboard

CATALOGUE

+ Add Category

Manage Categories

+ Add Author

Manage Authors

+ Add Book

Manage Books

CIRCULATION

Issue New Book

Manage Issued Books

BOOK STORE

Book Orders 1

STUDENTS

Registered Students

Search Student

All Orders (2)

Filter by order number, student, ID...

#	ORDER NO.	STUDENT	STUDENT ID	ORDER DATE	BOOKS	TOTAL	PAYMENT METHOD	PAYMENT	STATUS	ACTION
1	ORD100002	Priya Patel	SID10002	24 Sep 2026, 08:11 PM	2	Rs. 1,349.00	Pay at Library Counter	Unpaid	Pending	Manage
2	ORD100001	Rahul Sharma	SID10001	15 Sep 2026, 08:11 PM	2	Rs. 900.00	Cash on Delivery	Paid	Delivered	Manage

© 2026 Online Library Management System – Admin Panel

Figure 6.15: Book orders (admin)

Admin Panel

Student Details

Administrator Librarian / Admin

Dashboard

CATALOGUE

+ Add Category

Manage Categories

+ Add Author

Manage Authors

+ Add Book

Manage Books

CIRCULATION

Issue New Book

Manage Issued Books

BOOK STORE

Book Orders 1

STUDENTS

Registered Students

Search Student

Student Profile

Issue Book to this Student Back

Student ID: SID10001

Full Name: Rahul Sharma

Email: rahul@example.com

Mobile: 9876543210

Status: Active

Registered On: 25 Sep 2026, 08:11 PM

Last Updated: -

Books Pending: 1

Total Fine Paid: Rs. 10.00

Book Issue History (2)

#	BOOK	AUTHOR	ISBN	ISSUED ON	DUE DATE	RETURNED ON	FINE (RS.)	STATUS	ACTION
1	Clean Code	Robert C. Martin	9780132350884	20 Sep 2026, 08:11 PM	04 Oct 2026	-	0.00	Not Returned	Return
2	Hamlet	William Shakespeare	9780743477123	26 Aug 2026, 08:11 PM	09 Sep 2026	11 Sep 2026, 08:11 PM	10.00	Returned	View

Figure 6.16: Student details (admin)

7 Software Testing & Test Cases

7.1 Testing Strategy

Testing checks the main user journeys with valid inputs, invalid inputs and boundary conditions.

- **Unit testing:** single functions such as fine calculation, Student ID generation and image checks.
- **Integration testing:** modules working together, for example issue book and the student's issued books page.
- **System testing:** the complete flow from registration to issue, return, cart, checkout and order delivery.

All test cases in this chapter were run on the XAMPP setup. 94 automated checks covering every page and AJAX endpoint passed with no PHP errors.

7 Software Testing & Test Cases

7.2 Authentication Test Cases

Verify registration, login, password recovery, logout and access control.

TC ID	Test	Expected Result	Status
TC-01	Register with valid details	Account created, Student ID shown	Pass
TC-02	Register with an existing email	Error: email already registered	Pass
TC-03	Login with wrong password	Error: invalid ID or password	Pass
TC-04	Login as blocked student	Error: account blocked	Pass
TC-05	Forgot password with wrong mobile	Error: no account found	Pass
TC-06	Open admin page without login	Redirect to admin login	Pass

7 Software Testing & Test Cases

7.3 Book and Cart Test Cases

Verify book listing, live search and cart rules.

TC ID	Test	Expected Result	Status
TC-07	Search 'hamlet' in Listed Books	Only Hamlet card shown	Pass
TC-08	Add 3 copies when only 2 in stock	Error: only 2 copies available	Pass
TC-09	Add an out-of-stock book	Error: out of stock	Pass
TC-10	Change cart quantity to 3	Line and grand total updated	Pass
TC-11	Set quantity above 5	Error: choose 1 to 5	Pass
TC-12	Remove a book from cart	Book removed, total updated	Pass

7 Software Testing & Test Cases

7.4 Checkout and Order Test Cases

Verify checkout, stock update and order cancellation.

TC ID	Test	Expected Result	Status
TC-13	Cash on Delivery without address	Error: enter delivery address	Pass
TC-14	Place order with valid data	Order number shown, cart emptied	Pass
TC-15	Check stock after order	Stock reduced by ordered quantity	Pass
TC-16	Student cancels pending order	Order cancelled, stock restored	Pass
TC-17	Cancel the same order again	Error: cannot be cancelled	Pass
TC-18	Open another student's order	Error: order not found	Pass

7 Software Testing & Test Cases

7.5 Admin and Issue Test Cases

Verify catalogue management, issue/return and order processing.

TC ID	Test	Expected Result	Status
TC-19	Add duplicate category / ISBN	Error: already exists	Pass
TC-20	Delete category that has books	Error: cannot delete	Pass
TC-21	Upload a non-image file as cover	Error: upload a valid image	Pass
TC-22	Issue a book that is already issued	Error: already issued	Pass
TC-23	Return overdue book (6 days late)	Fine Rs. 30.00 calculated	Pass
TC-24	Set order status to Delivered	Payment marked Paid, order locked	Pass

7 Software Testing & Test Cases

7.6 Test Case Table

The following sample cases can be used during project demonstration.

TC ID	Module	Test	Expected Result
TC-01	Register	Valid student details	Account and Student ID created
TC-02	Login	Valid credentials	Session created, dashboard opens
TC-03	Books	Live search	Matching books shown
TC-04	Issue	Valid Student ID + ISBN	Book issued with return date
TC-05	Return	Return late book	Fine calculated and saved
TC-06	Cart	Change quantity	Total updated
TC-07	Checkout	Valid order data	Order created, stock reduced
TC-08	Admin	Update order status	Status saved
TC-09	Book	Add book with picture	Cover shown in all lists
TC-10	Security	Unauthorised admin access	Access restricted

8 Future Scope of Enhancements

8.1 Enhancement Opportunities

Future versions can add online payment (UPI / card), email and SMS reminders before the due date, barcode or QR code scanning for books, book reservation, e-books, reports in PDF/Excel, multiple copies per book with copy numbers, and a mobile app for students.

8 Future Scope of Enhancements

8.2 Security and Scalability

Before public use, the system should use HTTPS, CSRF protection, stronger password rules, login attempt limits, regular database backups and activity logs. For more users, it can use better indexing, caching and cloud hosting.

9 References

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4. MySQL Documentation – <https://dev.mysql.com/doc/>
5. jQuery API Documentation – <https://api.jquery.com/>
6. MDN Web Docs (HTML, CSS, JavaScript) – <https://developer.mozilla.org/>
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8. PHPGurukul, Online Library Management System (feature reference) – <https://phpgurukul.com/online-library-management-system/>

10 Appendix

A. Project Setup and Demonstration Steps

1. Install XAMPP (the installer is in the “xampp setup” folder of the project).
2. Start Apache and MySQL in the XAMPP Control Panel.
3. Copy the “library” folder to C:\xampp\htdocs\.
4. Open <http://localhost/phpmyadmin> and import library/database/library_db.sql.
5. Check database settings in library/includes/config.php (default: root, no password).
6. Open <http://localhost/library/> (student) or <http://localhost/library/admin/> (admin).
7. Use the demo accounts only for academic testing and change them before any public use.

B. Demo Accounts

Role	Username / Student ID	Password
Admin	admin	Password@123
Student	SID10001	Student@123
Student	SID10002	Student@123